AP CALCULUS AB

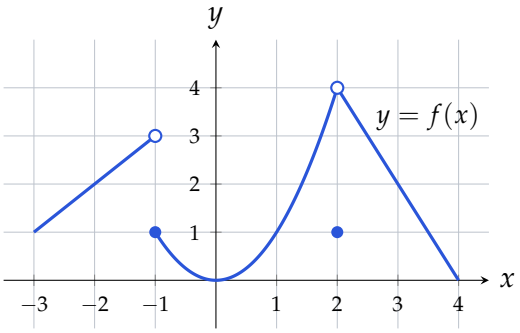
Unit 1 • Lesson 1.9 practice • Answer key

Lesson 1.9 • Connecting Multiple Representations of Limits

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.



The graph of f: a line segment for $-3 \le x < -1$, $y = x^2$ for $-1 \le x < 2$, a line segment for $2 < x \le 4$; $f(-1) = 1$, $f(2) = 1$.

Question 1

NO CALCULATOR

Let $g(x) = x + 1$. Using the graph of f, what is $\lim_{x \to 2} f(x)g(x)$?

(A) 3
(B) 4
(C) 12
(D) 7

Solution

Answer: (C). $\lim f(x) = 4$ and $\lim g(x) = 3$, so the product's limit is 12.

Question 2

NO CALCULATOR

With $g(x) = x + 1$, what is $\lim_{x \to 2} \big(f(x) - g(x)\big)$?

(A) 1
(B) -2
(C) -1
(D) 7

Solution

Answer: (A). $4 - 3 = 1$.

Question 3

NO CALCULATOR

A function h has values 4.9, 4.99, 4.999, 5.001, 5.01, 5.1 at $x = 0.9, 0.99, 0.999, 1.001, 1.01, 1.1$. Let $k(x) = 2x$. What is $\lim_{x\to 1} \dfrac{h(x)}{k(x)}$?

(A) 5
(B) $\dfrac{5}{2}$
(C) 10
(D) $\dfrac{2}{5}$

Solution
Answer: (B). The table suggests $\lim h(x) = 5$, and $\lim k(x) = 2$, so the limit is $\frac{5}{2}$.

Question 4

NO CALCULATOR

With $g(x) = x + 1$, what is $\lim_{x\to 3} g(f(x))$?

(A) 4
(B) 2
(C) 5
(D) 3

Solution
Answer: (D). $\lim_{x\to 3} f(x) = 2$ and g is continuous, so the limit is $g(2) = 3$.

Question 5 (graph)

NO CALCULATOR

Use the graph of f above. What is $\lim_{x\to -1^-} x\,f(x)$?

(A) -1
(B) 3
(C) -3
(D) It does not exist.

Solution
Answer: (C). $\lim_{x\to -1^-} x = -1$ and $\lim_{x\to -1^-} f(x) = 3$, so the limit is -3.

Question 6 (Free response)

NO CALCULATOR

Use the graph of f above. The function g is defined for all x, and $g(1.9) = -1.2$, $g(1.99) = -1.02$, $g(1.999) = -1.002$, $g(2.001) = -0.998$, $g(2.01) = -0.98$, $g(2.1) = -0.8$.

(a) Assuming the table shows the trend of g, find $\lim_{x\to 2} \big(f(x) + g(x)\big)$.
(b) Find $\lim_{x\to 2} f(x)g(x)$ and $\lim_{x\to 2} \dfrac{f(x)}{g(x)}$.
(c) Find $\lim_{x\to -1^+} \big(f(x) + x\big)$ and $\lim_{x\to -1^-} \big(f(x) + x\big)$.

Solution
(a) $\lim_{x\to 2} f(x) = 4$ (graph) and $\lim_{x\to 2} g(x) = -1$ (table), so the limit is 3.
(b) Product: $4 \cdot (-1) = -4$. Quotient: $\frac{4}{-1} = -4$ (the denominator's limit is not zero).
(c) Right: $1 + (-1) = 0$. Left: $3 + (-1) = 2$.